



R2.1 How much? The amount of chemical change

Reactivity 2 · How much, how fast and how far? · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A balanced chemical equation gives the stoichiometric mole ratio of reactants and products, which is used to calculate amounts, masses or volumes involved in a reaction; the limiting reagent is the reactant that is completely used up first and therefore determines the maximum theoretical amount of product (the reactant in excess remains partly unreacted). Percentage yield compares the actual mass of product obtained experimentally with the theoretical (maximum possible) yield calculated from stoichiometry, and is always less than 100% due to factors such as incomplete reactions, side reactions, and losses during purification. Volumetric analysis (titration) uses the stoichiometric mole ratio and the relationship $n = cV$ to determine an unknown concentration or amount from a known, accurately measured reacting volume.

Worked example: follow the reasoning

In a titration, 25.0 cm³ of a sodium hydroxide solution of unknown concentration is exactly neutralised by 21.5 cm³ of 0.100 mol dm⁻³ hydrochloric acid, according to the equation $\text{NaOH(aq)} + \text{HCl(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$. Calculate the concentration of the NaOH solution.

Step 1: $n(\text{HCl}) = c \times V = 0.100 \text{ mol dm}^{-3} \times (21.5/1000) \text{ dm}^3 = 0.00215 \text{ mol}$.

Step 2: From the 1:1 stoichiometry, $n(\text{NaOH}) = n(\text{HCl}) = 0.00215 \text{ mol}$.

Step 3: $c(\text{NaOH}) = n / V = 0.00215 \text{ mol} / (25.0/1000) \text{ dm}^3 = 0.00215 / 0.0250$.

Step 4: $c(\text{NaOH}) = 0.0860 \text{ mol dm}^{-3}$.

Answer: $[\text{NaOH}] = 0.0860 \text{ mol dm}^{-3}$.

Quick check · multiple choice

5.00 g of Zn ($M = 65.4 \text{ g mol}^{-1}$) reacts with 5.00 g of S ($M = 32.1 \text{ g mol}^{-1}$) to form ZnS according to $\text{Zn} + \text{S} \rightarrow \text{ZnS}$. Which reactant is in excess?

A Zn, because it has a greater mass

B S, because $n(\text{S}) > n(\text{Zn})$

C Zn, because $n(\text{Zn}) < n(\text{S})$

D Neither; both are used up exactly

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Calcium carbonate reacts with hydrochloric acid according to: $\text{CaCO}_3(\text{s}) + 2\text{HCl(aq)} \rightarrow \text{CaCl}_2(\text{aq}) + \text{H}_2\text{O(l)} + \text{CO}_2(\text{g})$. [8]
- A student reacts 2.50 g of CaCO_3 ($M = 100.1 \text{ g mol}^{-1}$) with 50.0 cm³ of 1.00 mol dm⁻³ HCl.
- (a) Show, by calculation, which reactant is the limiting reagent. [3]
- (b) Calculate the theoretical (maximum) mass of CaCl_2 ($M = 111.0 \text{ g mol}^{-1}$) that could be formed. [2]
- (c) The student actually obtains 2.60 g of CaCl_2 after the reaction and evaporation. Calculate the percentage yield. [2]
- (d) Suggest one reason why the percentage yield in (c) is less than 100%. [1]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) $n(\text{CaCO}_3) = 2.50/100.1 = 0.0250 \text{ mol}$; requires $2 \times 0.0250 = 0.0500 \text{ mol HCl}$ to react completely. $n(\text{HCl}) \text{ available} = 1.00 \times (50.0/1000) = 0.0500 \text{ mol}$. Since exactly 0.0500 mol HCl is available and exactly 0.0500 mol is required, neither is technically in excess for this dataset — accept: they react in exactly the stoichiometric ratio, so there is no single limiting reagent (both are fully consumed together). [3: correct moles of each, correct required ratio, correct conclusion]

(b) $n(\text{CaCl}_2) = n(\text{CaCO}_3) = 0.0250 \text{ mol}$ (1:1 ratio). $\text{Mass} = 0.0250 \times 111.0 = 2.775 \text{ g} \approx 2.78 \text{ g}$. [2]

(c) $\% \text{ yield} = (\text{actual/theoretical}) \times 100 = (2.60/2.775) \times 100 = 93.7\%$. [2]

(d) Any valid reason, e.g. some product was lost during transfer/filtration/evaporation, or the reaction did not go to completion, or some CaCl_2 remained in solution and was not fully recovered. [1]

Quick check answer

B. $n(\text{Zn}) = 5.00/65.4 = 0.0765 \text{ mol}$; $n(\text{S}) = 5.00/32.1 = 0.156 \text{ mol}$. Since the equation requires a 1:1 ratio, S (0.156 mol) is present in a greater amount than needed to react with all the Zn, so S is in excess.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.